IMMUNITY TO MICROBES

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2nd Stage

IBSII / LG 6

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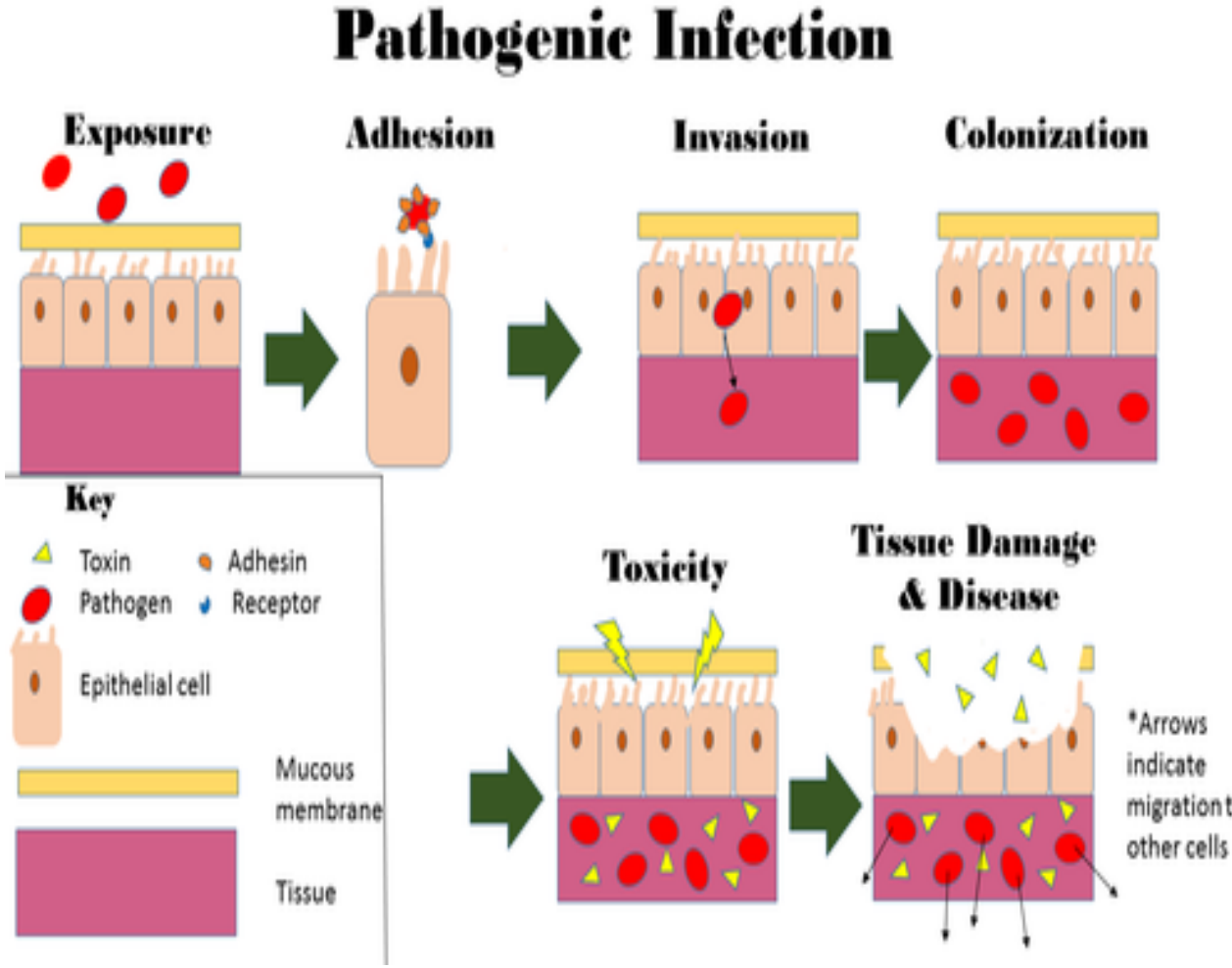
I. INTRODUCTION AND GENERAL PRINCIPLES

- Most infectious diseases are caused by pathogens much smaller than a human cell.
- For these microbes, the human body constitutes a vast, resource-rich environment in which to live and reproduce.
- Without a functional immune system, people would quickly succumb to the countless infectious microorganisms in the environment.

I. INTRODUCTION AND GENERAL PRINCIPLES

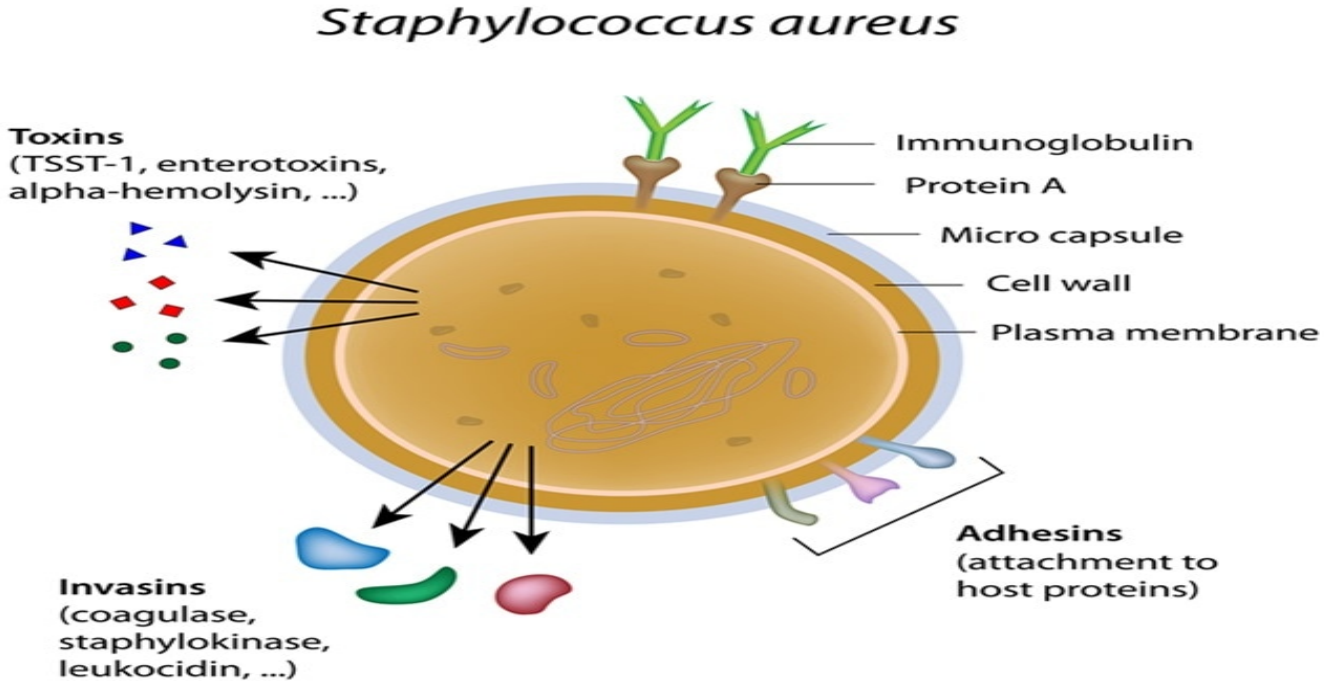
- The development of an infectious disease involves complex interactions between the microbe and the host.

- The key events during infection include:
- Entry of the microbe
- Invasion and colonization of host tissues
- Evasion of host immunity
- Tissue injury or functional impairment



I. INTRODUCTION AND GENERAL PRINCIPLES

- Many features of microorganisms determine their virulence, and many diverse mechanisms contribute to the pathogenesis of infectious diseases.



Infectious diseases are caused by six types of pathogens:

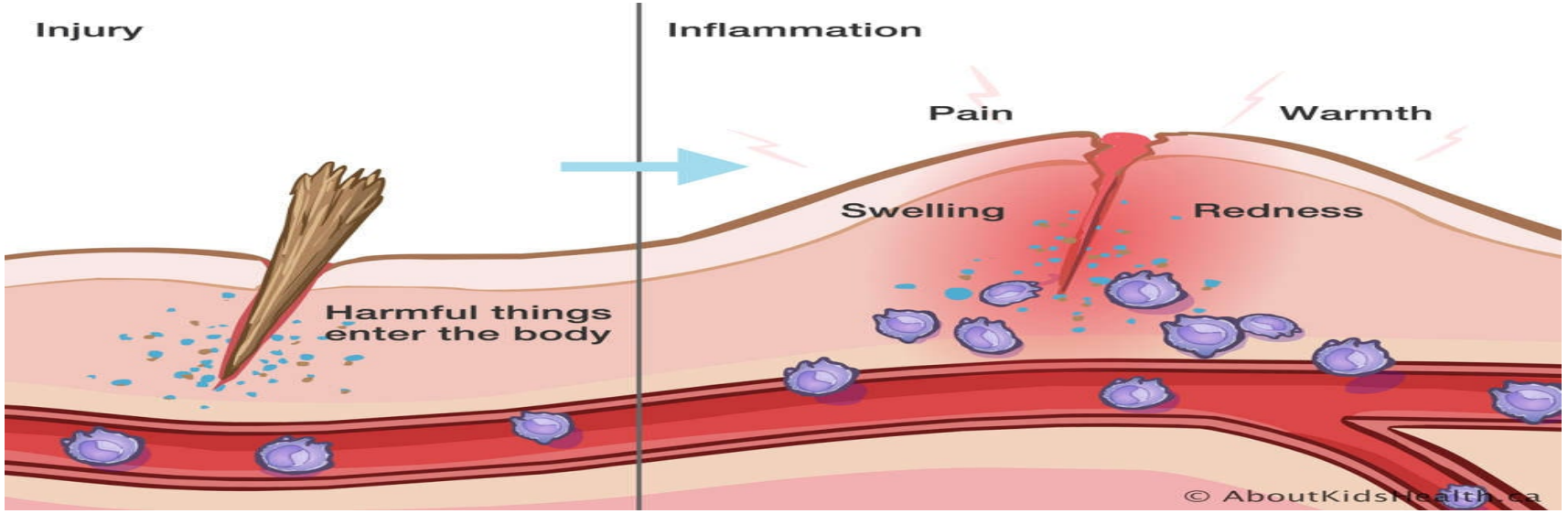
Pathogen Type	Key Characteristics
Extracellular bacteria	Do not need to enter host cells to reproduce
Intracellular bacteria	Must enter host cells to reproduce
Viruses	Submicroscopic, acellular particles; must enter host cells to replicate
Parasites	Eukaryotic organisms; may be protozoans, helminths, or arthropods
Fungi	Eukaryotic organisms; can exist outside a host but invade if given the opportunity
Prions	Infectious proteins that cause neurological disease

The Two Arms of Anti-Microbial Defense

- Defense against microbes is mediated by the effector mechanisms of **innate and adaptive immunity**:
- **Innate immunity** provides early defense. Its mechanisms can eliminate most microbes without activating adaptive immunity.
- **Adaptive immunity** is critically important for protection against pathogenic microbes that have evolved to resist innate immunity.
- In adaptive responses, large numbers of effector cells and antibody molecules are generated to eliminate microbes, along with memory cells that protect against repeated infections.

II. IMMUNITY TO EXTRACELLULAR BACTERIA

- **Disease Mechanisms**
- Many different species of extracellular bacteria are pathogenic. They induce disease by two principal mechanisms:
- **Inflammation:** Results in tissue destruction at the site of infection.



II. IMMUNITY TO EXTRACELLULAR BACTERIA

- **Disease Mechanisms**
- **Toxins:** Pathogenic bacteria produce toxins with diverse pathologic effects.
  - **Endotoxins:** Components of cell walls released by dying Gram-negative bacteria. Lipopolysaccharide (LPS) is a potent activator of macrophages, dendritic cells, and endothelial cells.
  - **Exotoxins:** Secreted by living bacteria. Examples include diphtheria toxin (shuts down protein synthesis), cholera toxin (interferes with ion/water transport), tetanus toxin (inhibits neuromuscular transmission), and anthrax toxin (disrupts signaling pathways).

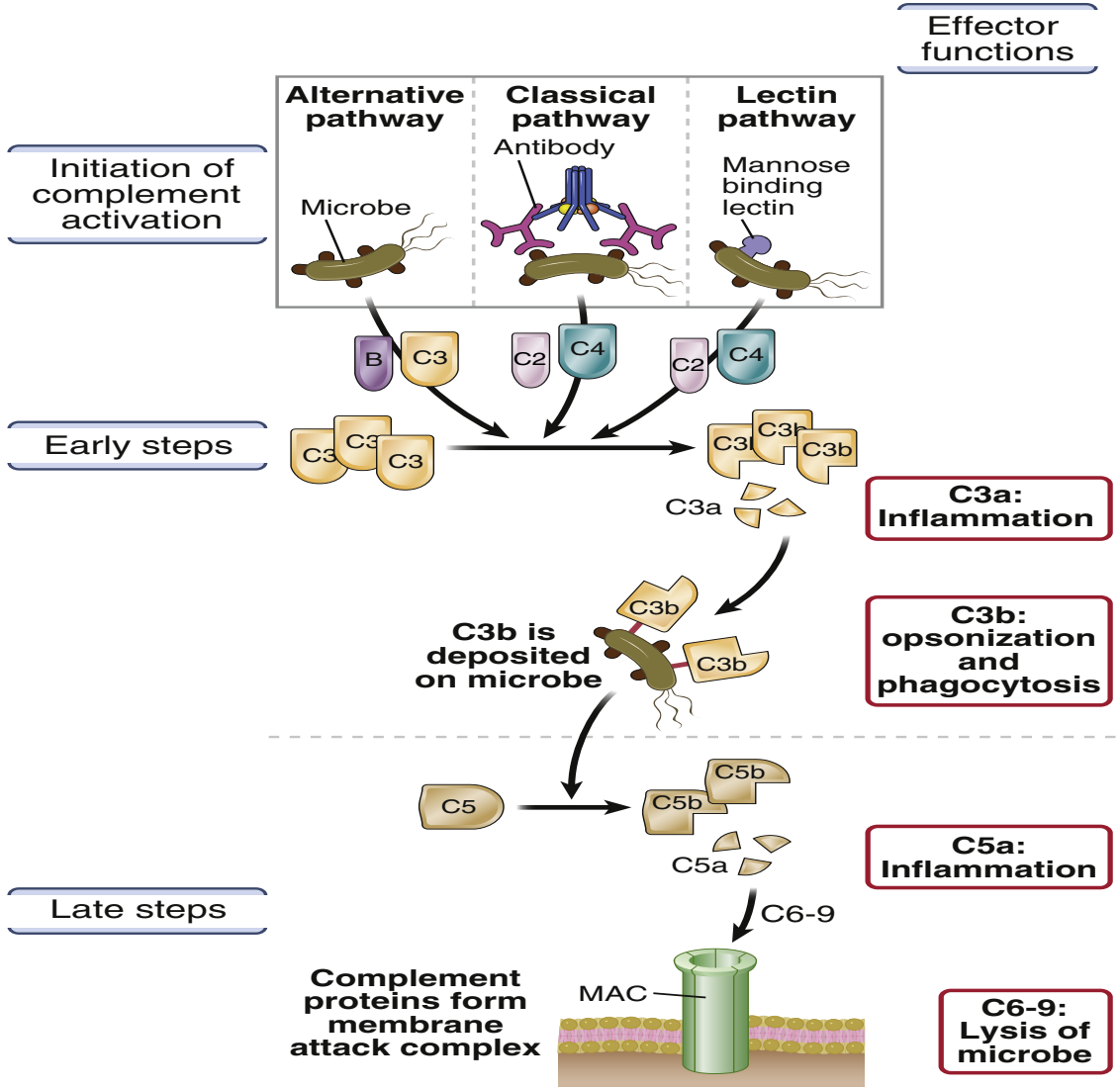
IMMUNITY TO EXTRACELLULAR BACTERIA
Innate Immune Effector Mechanisms

- The principal mechanisms of innate immunity to extracellular bacteria are:
- Complement activation
- Phagocytosis
- Inflammatory responses

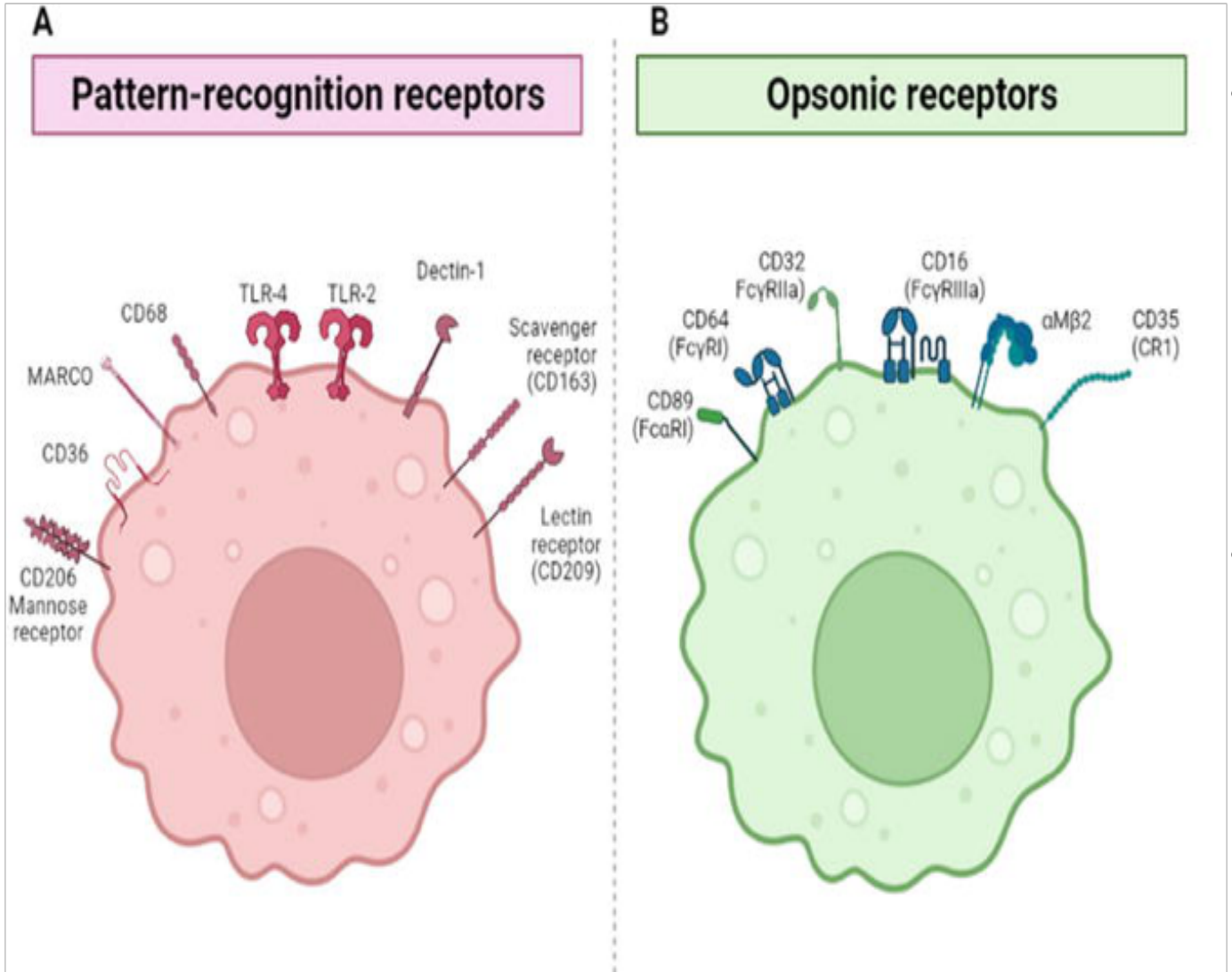
- **Complement Activation Pathways:**
- **Alternative pathway:** Activated by peptidoglycans (Gram-positive) and LPS (Gram-negative).
- **Lectin pathway:** Mannose-binding lectin (MBL) binds mannose on bacterial surfaces.
- **Classical pathway:** C-reactive protein binds phosphorylcholine on bacterial surfaces.
- **Consequences of Complement Activation:**
- Opsonization and enhanced phagocytosis
- Membrane attack complex lyses bacteria (especially *Neisseria* species)
- C3a and C5a stimulate inflammatory responses by recruiting and activating leukocytes

Pathways of complement activation.

- The activation of the complement system (the early steps) may be initiated by three distinct pathways, all of which lead to the production of C3b.
- C3b initiates the late steps of complement activation, culminating in the formation of a multiprotein complex called the mem- brane attack complex *(MAC),* which is a transmembrane channel composed of polymerized C9 molecules that causes lysis of thin-walled microbes.
- Peptide by-products released during complement activation are the inflammation-inducing C3a and C5a.

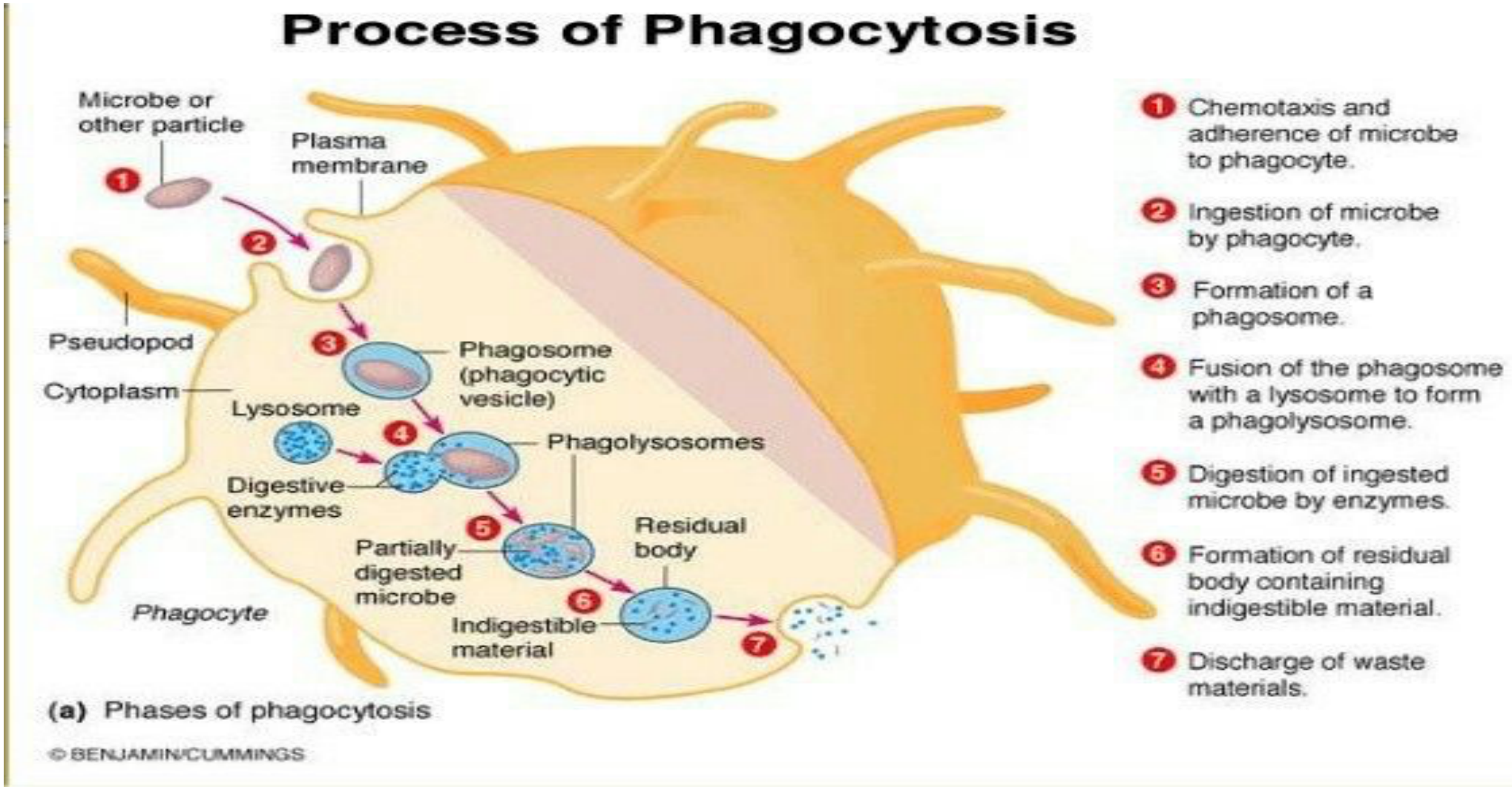


Phagocytosis: Neutrophils and macrophages use surface receptors (mannose receptors, scavenger receptors) to recognize extracellular bacteria.



- Phagocytic receptors are present in the
- A) Pattern recognition receptors (PRR): TLR, scavenger receptors, lectin receptors, mannose receptors, which recognize pathogen-associated molecular patterns (PAMPs) and damage-associated molecular patterns (DAM).
- B) Opsonic receptors: such as receptors for the crystallizable fraction of antibodies (FcR), complement receptors (CR), which recognize antibodies and C1 or C3b molecules that opsonize microorganisms and promote phagocytosis.

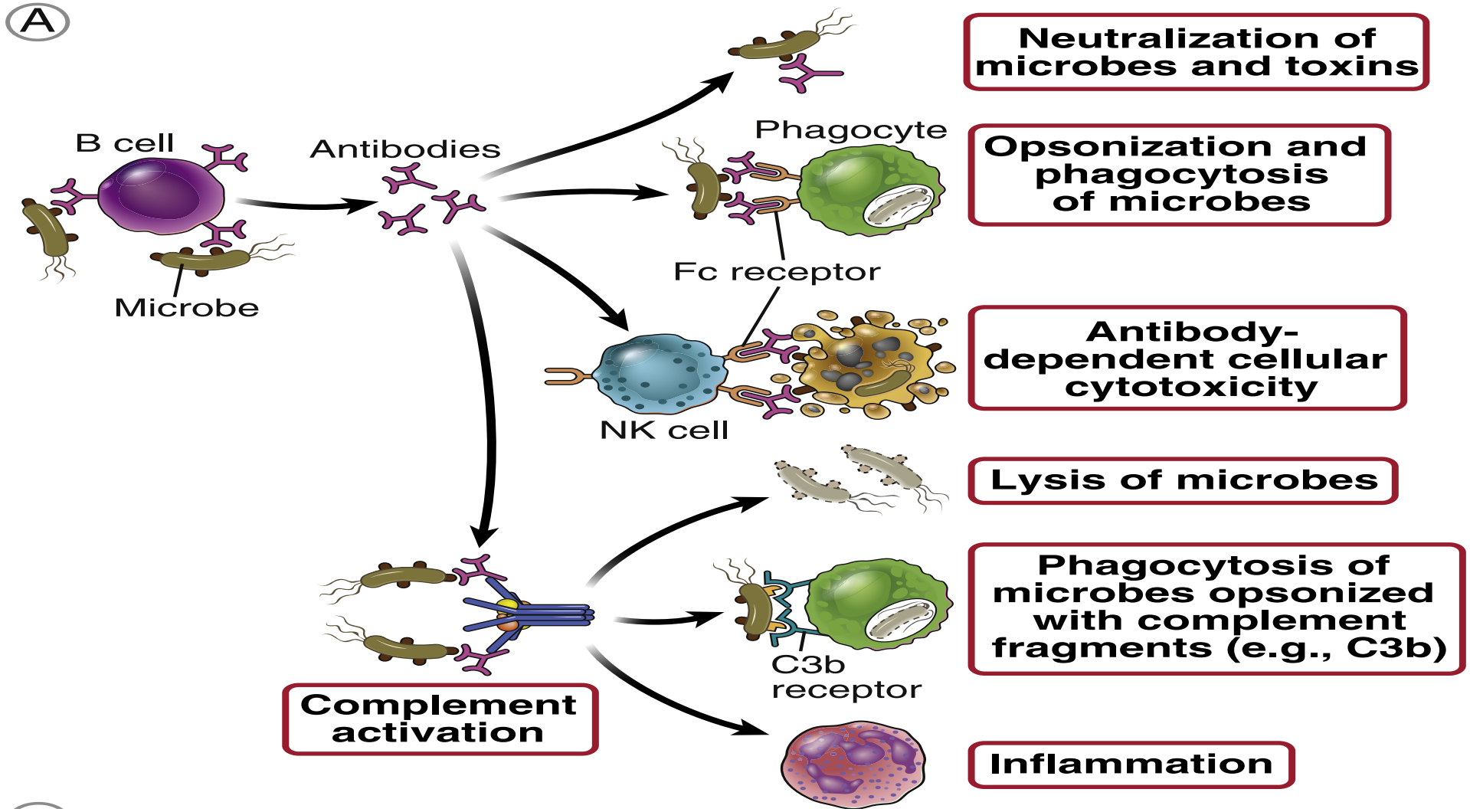
Phagocytosis



IMMUNITY TO EXTRACELLULAR BACTERIA
Adaptive Immune Effector Mechanisms

- **Humoral immunity:** Antibodies (especially IgG) opsonize bacteria, neutralize toxins, and activate complement.
- **Th17 responses:** Promote inflammation and recruit neutrophils to infection sites.
- **Th1 responses:** Support macrophage activation for killing of phagocytosed bacteria.

Effector functions of antibodies. Antibodies are produced by the activation of B lymphocytes by antigens and other signals(against extracellular bacterial pathogens)

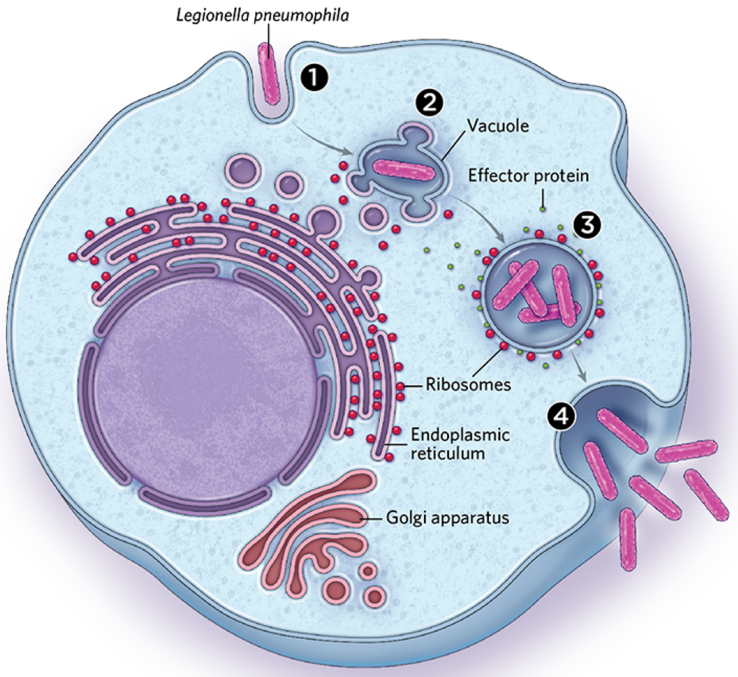


III. IMMUNITY TO INTRACELLULAR BACTERIA

- **Disease Mechanisms**
- Intracellular bacteria have evolved to survive and replicate within host cells, particularly macrophages. They cause disease by:
- Directly damaging infected cells
- Inducing chronic inflammation and granuloma formation
- Evading killing mechanisms within phagocytes

IMMUNITY TO INTRACELLULAR BACTERIA

- ***Legionella* takes control of host membranes**
  Some intracellular bacteria, such as *Legionella pneumophila*, inhabit membrane-bound compartments inside host cells



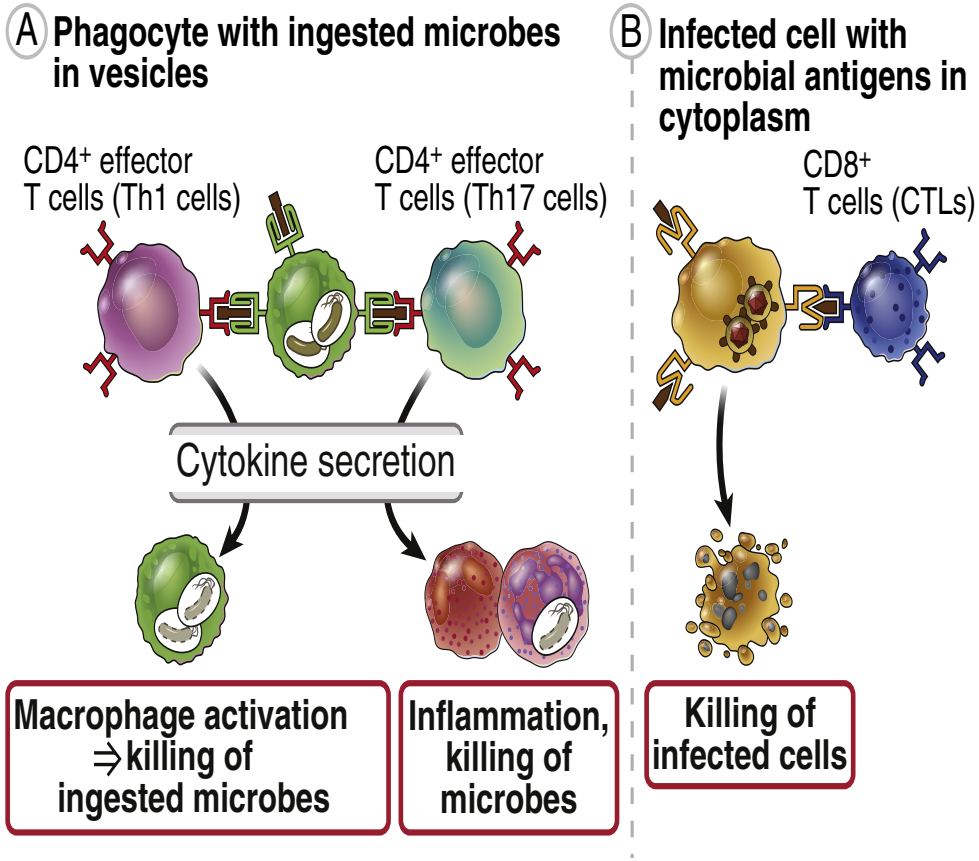
IMMUNITY TO INTRACELLULAR BACTERIA

- **Innate Immune Effector Mechanisms**
- **NK cells:** Early production of IFN-γ to activate macrophages
- **Macrophages:** Attempt to kill bacteria via reactive oxygen species, nitric oxide, and lysosomal enzymes
- **γδ T cells and NKT cells:** Contribute to early cytokine production

IMMUNITY TO INTRACELLULAR BACTERIA

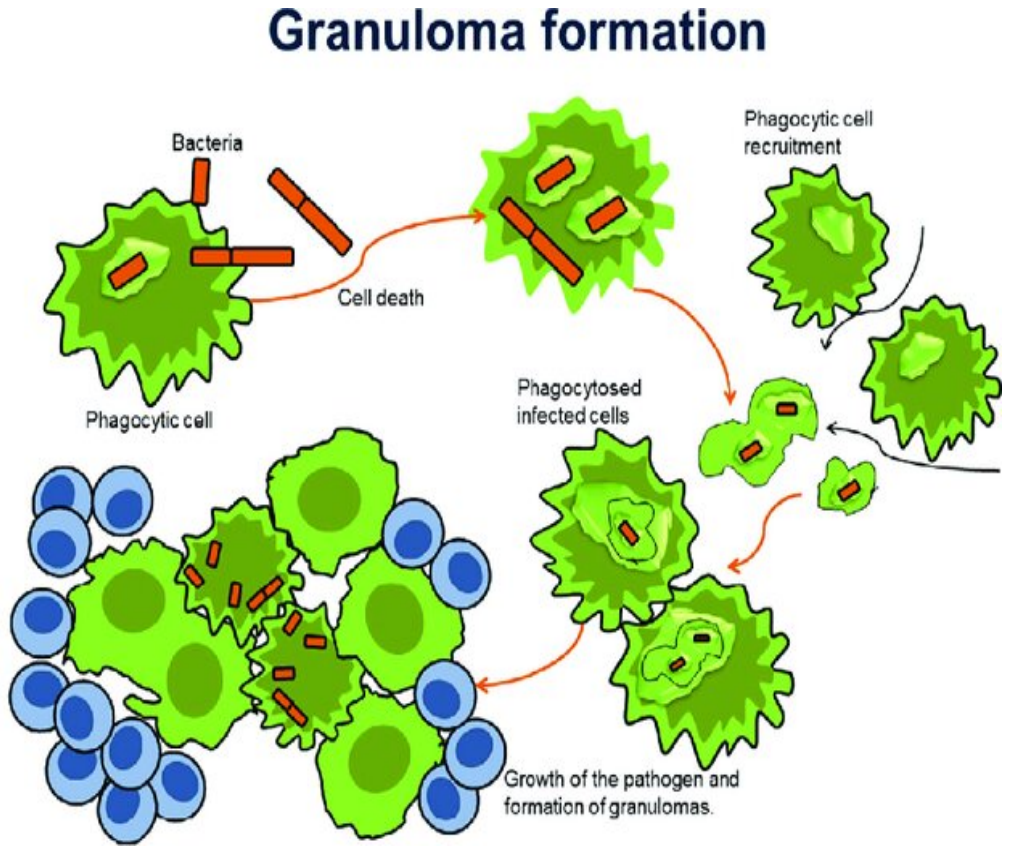
- **Adaptive Immune Effector Mechanisms**
- **Th1 responses (critical):** $CD4^+$ Th1 cells secrete **IFN-γ**, which activates macrophages to enhance their microbicidal activity.
- **$CD8^+$ cytotoxic T cells:** Kill infected cells that present bacterial antigens on MHC class I.
- **Granuloma formation:** Macrophage hyperactivation and granuloma formation confine persistent invaders

- Cell-mediated Immunity. A, Effector T cells of the CD4+ Th1 and Th17 subsets recognize microbial antigens and secrete cytokines that recruit leukocytes (inflammation) and activate phagocytes to kill the microbes
- B, CD8+ cytotoxic T lymphocytes (CTLs) kill infected cells with microbial antigens in the cytosol. CD8+ T cells also produce cytokines that induce inflammation and activate macrophages.



Granuloma formation

Definition	Chronic inflammation with macrophages, epithelioid histiocytes, and giant cells
Core Cell Type	Activated macrophages (epithelioid cells)
Immune Response	Th1 → IFN-γ → macrophage activation (Type IV hypersensitivity)
Purpose	Wall off a persistent antigen or pathogen



IV. IMMUNITY TO VIRUSES

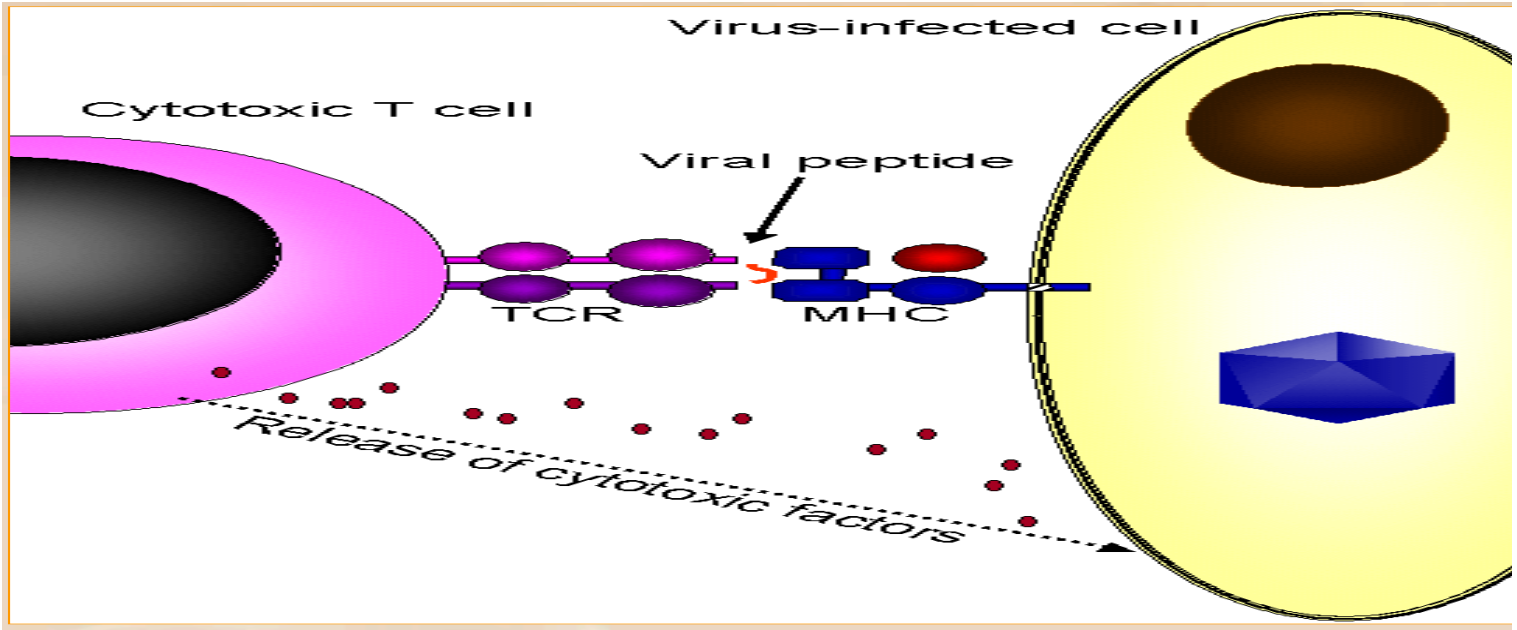
- **Disease Mechanisms**
- Viruses are submicroscopic, acellular particles that must enter a host cell and exploit its protein synthesis machinery to propagate. They cause disease by:
- Directly killing infected cells (cytopathic effect)
- Inducing immune-mediated damage
- Transforming cells (oncogenesis)

IMMUNITY TO VIRUSES

- **Innate Immune Effector Mechanisms**
- **Type I interferons (IFN-α/β):** Critical for inducing an antiviral state in uninfected cells
- **NK cells:** Kill virus-infected cells that downregulate MHC class I
- **Complement:** Can neutralize some viruses
- **γδ T cells and NKT cells:** Contribute to early defense

IMMUNITY TO VIRUSES

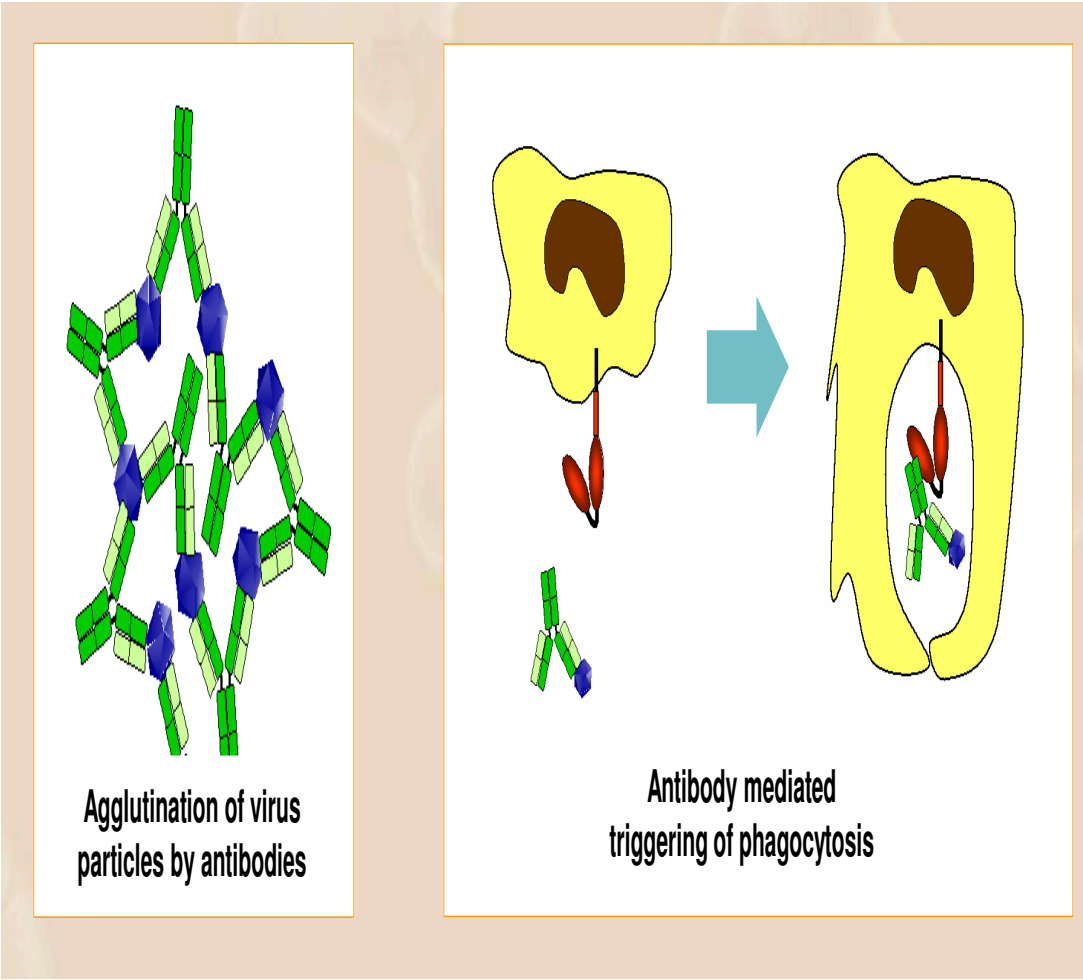
- **Adaptive Immune Effector Mechanisms**
- **$CD8^+$ cytotoxic T cells (CTLs):** Eliminate virus-infected cells by recognizing viral peptides on MHC class I.
- **$CD4^+$ Th1 cells:** Provide help for CTL responses and activate macrophages.
- **Antibodies:** Neutralize free virus particles, preventing infection of new cells. Secreted IgA blocks attachment at mucosal surfaces.



Viruses can also be removed from the body by antibodies before they get the chance to infect a cell.

- Firstly, the antibodies neutralise the virus, meaning that it is no longer capable of infecting the host cell.
  • Secondly, many antibodies can work together, causing virus particles to stick together in a process called agglutination.
- Agglutinated viruses make an easier target for immune cells than single viral particles.

- A third mechanism used by antibodies to eradicate viruses, is the activation of phagocytes. A virus-bound antibody binds to receptors, called Fc receptors, on the surface of phagocytic cells and triggers phagocytosis, by which the cell engulfs and destroys the virus.
  • Finally, antibodies can also activate the complement system, which opsonises and promotes phagocytosis of viruses.
- Complement can also damage the envelope (phospholipid bilayer) that is present on some types of virus.



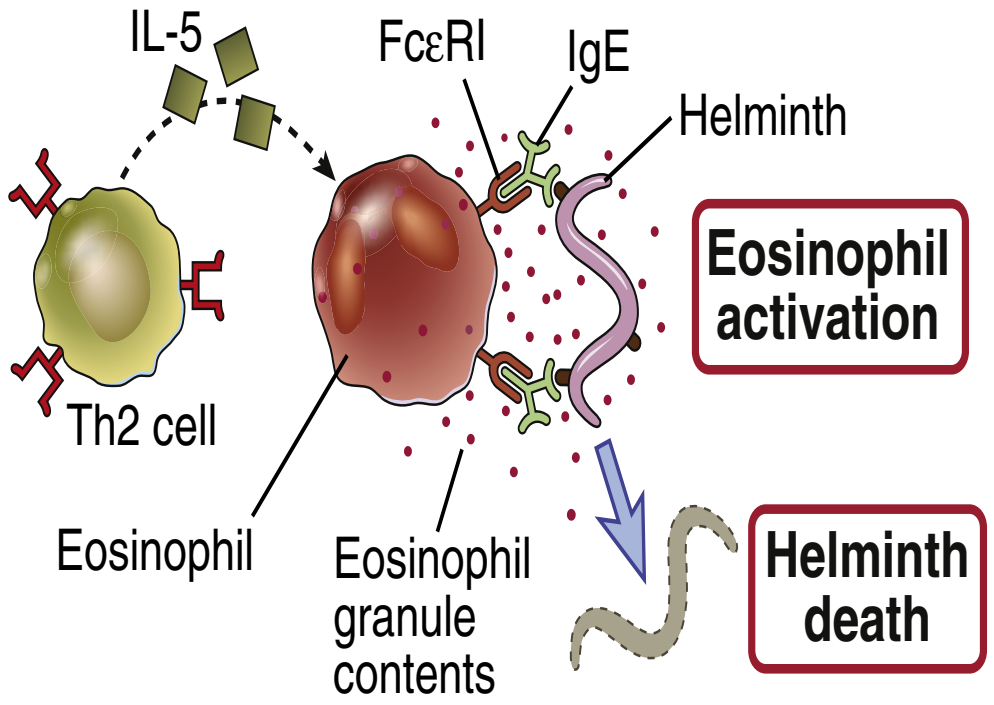
V. IMMUNITY TO PARASITES

- **Disease Mechanisms**
- Parasites are eukaryotic organisms that take advantage of a host for habitat and nutrition at some point in their life cycles. They often damage a host but kill it only slowly.
- **Innate Immune Effector Mechanisms**
- **Complement:** Can lyse some protozoa
- **Macrophages:** Phagocytose and kill protozoa
- **Eosinophils and mast cells:** Important for helminth defense
- **NK cells:** Provide early IFN-γ

IMMUNITY TO PARASITES

- **Adaptive Immune Effector Mechanisms**
- **Th2 responses (for helminths):** IL-4, IL-5, and IL-13 drive IgE production, eosinophil activation, and mast cell degranulation.
- **IgE and IgA:** Prevent parasitic worms from anchoring in the host.
- **IgE triggers mast cell, basophil, and eosinophil degranulation** with release of toxic mediators.
- **Th1 responses (for protozoa):** IFN-γ activates macrophages for intracellular killing.

Immune response against helminths



- Immunoglobulin E *(IgE)*- and eosinophil-mediated killing of helminths. IgE antibody binds to helminths and recruits and activates eosinophils via FcεRI, leading to degran- ulation of the cells and release of toxic mediators. Interleukin-5 *(IL-5)* secreted by Th2 cells enhances the ability of eosinophils to kill the parasites. *Ig,* Immunoglobulin.

VI. IMMUNITY TO FUNGI

- **Disease Mechanisms**
- Fungi are eukaryotic organisms that can exist comfortably outside a host but will invade and colonize that host if given the opportunity. They may be single-celled or multicellular. Most fungal infections occur in immunocompromised hosts.

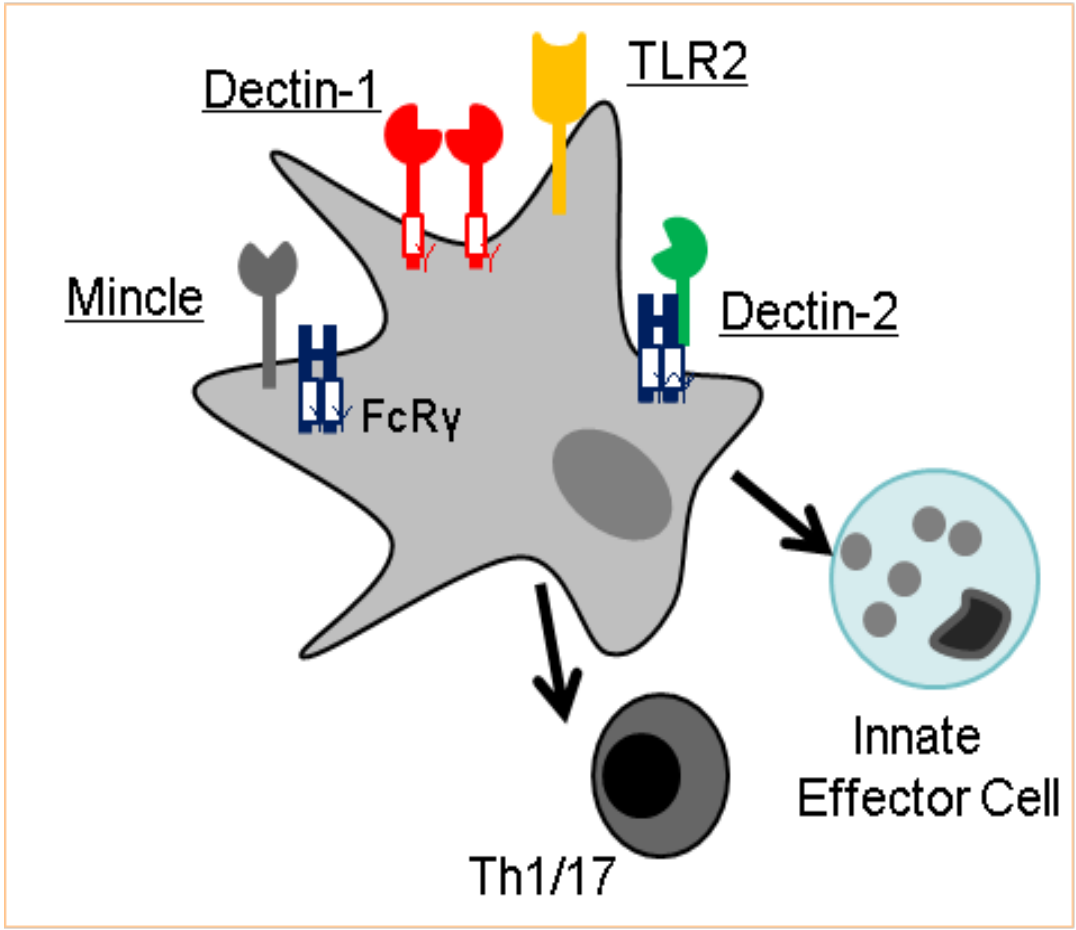
VI. IMMUNITY TO FUNGI

- **Innate Immune Effector Mechanisms**
- **Neutrophils:** Primary phagocytes for fungi
- **Macrophages:** Phagocytose and kill fungi
- **Complement:** Opsonizes fungi for phagocytosis
- **Antimicrobial peptides:** Defensins, histatins

- **Adaptive Immune Effector Mechanisms**
- **Th1 responses:** IFN-γ activates macrophages for enhanced antifungal activity
- **Th17 responses:** IL-17 recruits neutrophils to mucosal sites
- **Antibodies:** Opsonize fungi and activate complement

IMMUNITY TO FUNGI

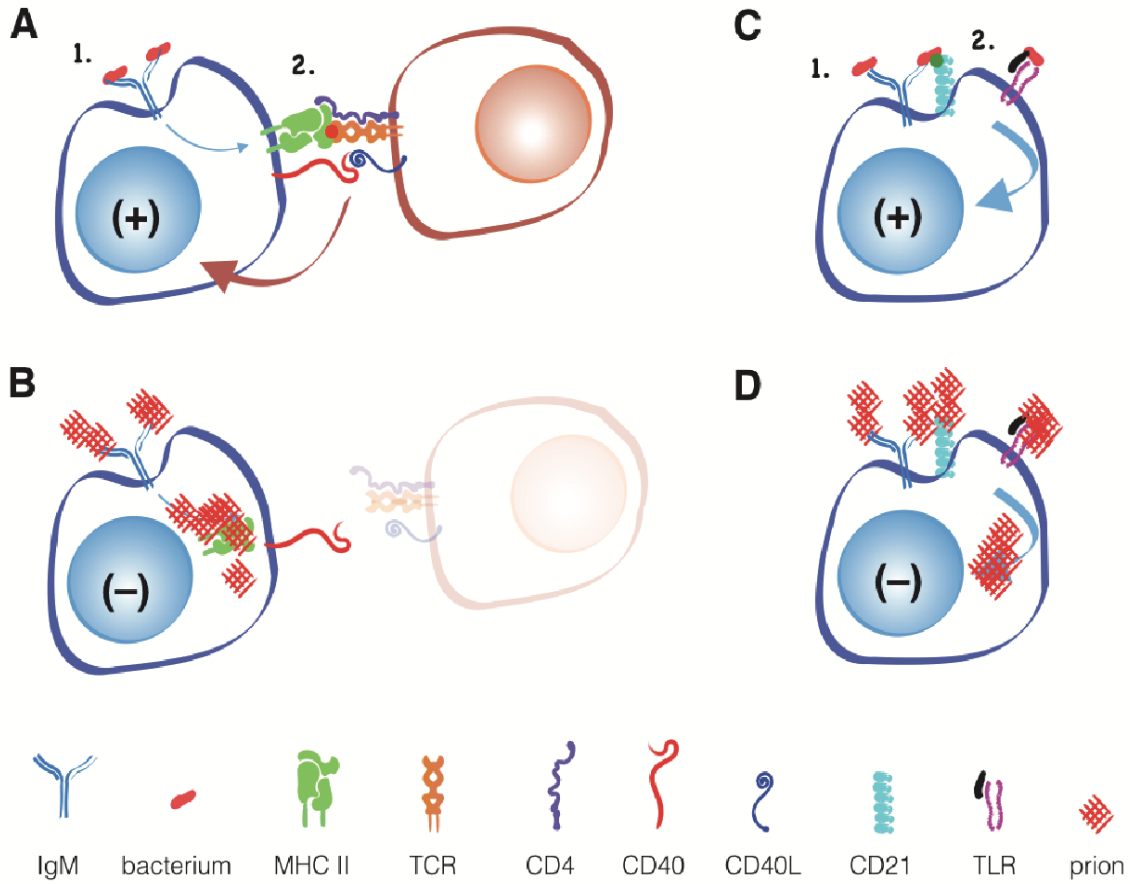
A dendritic cell decorated with anti-fungal PRRs. Upon recognition of fungi through these receptors, signalling results in generation of Th1/17 adaptive immune responses and activation of innate effector cells



VII. IMMUNITY TO PRIONS

- **Disease Mechanisms**
- Prions are infectious proteins that cause neurological disease by altering normal proteins in the brain of the infected host. They cause:
- **Transmissible spongiform encephalopathies** (e.g., Creutzfeldt-Jakob disease, kuru, BSE)
- **Neurodegeneration** without inflammatory response
- **Immune Response**
- **No effective immune response:** Prions are composed of normal host protein (**cellular prion protein** PrP^C) in an abnormal conformation (PrP^Sc), so they are recognized as "self."
- **No antibodies or T cell responses** are generated
- **No inflammation** is observed in affected brain tissue

- Prions fail to activate B cells with or without T cell help. A simplified model of T cell-dependent and independent B cell activation. Binding of antigen to IgM on the surface of B cells provides the first activation signal.
- (A) T cells that recognize antigen processed by the B cell and presented in MHC II molecules provide the second signal, typically through its CD40 ligand binding CD40 on the B cell during infections with classic pathogens.
- (B) Prion-specific T cells are absent during prion infection. Other innate immune receptors, like CD21 or TLRs, can provide secondary signals without T cells in response to
- (C) bacteria, but not to
- (D) prions.



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